Gap-Aware Retrieval (GAR)

A Gap-Aware Retrieval Algorithm for Information-Seeking RAG Systems

Abstract

This project proposes Gap-Aware Retrieval (GAR), a retrieval algorithm designed to retrieve information that is still missing from the current evidence rather than relying only on query-document similarity.

The project uses MIGRES as the base system. MIGRES already identifies missing information during iterative retrieval, but its retrieval component uses BM25. Our goal is to replace the retrieval component with GAR while keeping the rest of the MIGRES system unchanged.

GAR combines three ideas: relevance to the current information gap, coverage of the missing information, and novelty with respect to previously collected evidence. The algorithm is designed to work independently and can also be integrated into MIGRES or other retrieval-augmented systems.

The project will compare GAR against BM25, DPR, Contriever, ColBERTv2, and SPLADE inside the same MIGRES framework. Retrieval quality, evidence coverage, answer accuracy, latency, and computational cost will be evaluated. Ablation studies and statistical hypothesis testing will be used to determine whether each part of GAR contributes to its performance.

1 Project Overview

The main goal of this project is to design a new retrieval algorithm called:

Gap-Aware Retrieval (GAR)

The basic idea is simple:

> Instead of asking only "Which document is relevant to my question?", GAR asks "Which document helps fill the information that is still missing?"

Traditional retrieval systems usually focus on query-document similarity. This works well when the question can be answered directly from one retrieval step.

However, complex questions often require multiple pieces of evidence.

For example:

> **Question:** Why did Company X's revenue increase?

The system may already know:

> Company X's revenue increased by 30%.

The important missing information is:

> What caused the increase?

A traditional retriever may retrieve many documents that again mention the 30% increase.

GAR attempts to prefer documents that provide the missing information.

2 Base Paper: MIGRES

Our base paper is MIGRES.

MIGRES provides the main motivation for this project because it uses the concept of **missing information** during retrieval.

The general process is:

$$\text{Question} \rightarrow \text{Retrieve Evidence} \rightarrow \text{Identify Missing Information} \rightarrow \text{Retrieve Again}$$

The important observation for our project is that MIGRES uses BM25 as its retrieval component in its reported setup.

Therefore, instead of modifying the complete MIGRES system, we can keep the system fixed and replace only its retrieval algorithm.

The original setup can be represented as:

$$\text{MIGRES} + \text{BM25}$$

Our proposed setup is:

$$\text{MIGRES} + \text{GAR}$$

This creates a clean experimental comparison.

2.1 Base Paper

MIGRES — ACL Anthology:

https://aclanthology.org/2025.coling-main.163/

3 How the Research Idea Was Formed

The mathematical model is not copied from one paper.

Instead, the project starts from MIGRES and combines several related retrieval ideas.

Three ideas are especially important.

3.1 1. Missing Information

MIGRES gives us the idea of identifying information that is still missing.

We represent this missing information as:

$$G$$

where G is the current information gap.

3.2 2. Evidence Gap

The paper on Unsupervised Alignment-based Iterative Evidence Retrieval gives us the idea of examining what the current evidence does not cover.

This motivates the idea of measuring:

$$C(d, G)$$

which represents how much a candidate document covers the current information gap.

3.3 3. Information Gain

Relevant Information Gain and InfoGain-RAG motivate the idea that a useful document should add information instead of simply being relevant.

This motivates:

$$N(d, E)$$

where N measures how different or novel the candidate document is compared with the evidence already collected.

Therefore, the three central ideas become:

$$\boxed{\text{Gap Relevance} + \text{Gap Coverage} + \text{Evidence Novelty}}$$

These ideas are then formulated into one retrieval objective.

4 Simple Example

Consider the question:

> Why did Company X's revenue increase?

Suppose the current evidence says:

> Company X's revenue increased by 30%.

Therefore, the information gap is:

> What caused the increase?

Now consider three candidate documents.

Document A:

> Company X's revenue increased by 30%.

This is relevant but mostly repeats information that we already know.

Document B:

> Company X's revenue increased because demand for its AI products grew.

This directly addresses the information gap.

Document C:

> Company X's stock price increased by 20%.

This may be related to the company but does not answer the current information gap.

GAR should therefore prefer:

$$B > A > C$$

because Document B provides the most useful new information for resolving the gap.

5 GAR Mathematical Model

Let:

$$
\begin{aligned}
q &= \text{original question} \\
E &= \text{evidence already collected} \\
G &= \text{current information gap} \\
d &= \text{candidate document} \\
D &= \text{document corpus}
\end{aligned}
$$

The proposed GAR score is:

$$\boxed{S(d \mid q, E, G) = \alpha R(d, q, G) + \beta C(d, G) + \gamma N(d, E)}$$

The retrieval algorithm selects the documents with the highest score:

$$\boxed{TopK = \mathrm{TopK}_{d \in D}\, S(d \mid q, E, G)}$$

In simple English:

> GAR selects documents that are relevant to the question and missing information, cover the missing information, and provide information that is not already present in the evidence.

6 Term 1: Gap Relevance

The first component is:

$$R(d, q, G)$$

It asks:

> How relevant is document d to the original question and the current information gap?

A simple proposed formulation is:

$$\boxed{R(d, q, G) = \lambda \,\mathrm{sim}(d, q) + (1 - \lambda)\,\mathrm{sim}(d, G)}$$

Here:

- $\mathrm{sim}(d, q)$ measures similarity between the document and the original question.

- $\mathrm{sim}(d, G)$ measures similarity between the document and the information gap.

Similarity can be implemented using cosine similarity between embeddings.

For example:

$$\mathrm{sim}(d, q) = 0.8$$

and

$$\mathrm{sim}(d, G) = 0.9$$

would indicate that the document is strongly related to both the question and the missing information.

7 What is Similarity?

The function sim simply means **similarity**.

For two vectors $\mathbf{x}$ and $\mathbf{y}$, cosine similarity can be written as:

$$\boxed{\mathrm{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}}$$

If documents and queries are represented by embeddings, we can calculate their similarity using this function.

For example:

> Query: "cause of revenue increase"

may be highly similar to:

> Revenue increased because demand for AI products grew.

while it may have low similarity to:

> The company opened a new office in London.

8 Term 2: Gap Coverage

The second component is:

$$C(d, G)$$

It asks:

How much of the missing information does document d actually provide?

The information gap may contain multiple requirements:

$$G = \{g_1, g_2, \ldots, g_k\}$$

For example:

$$G = \{\text{cause of increase}, \text{supporting evidence}\}$$

A proposed coverage function is:

$$\boxed{C(d, G) = \frac{\sum_{i=1}^{k} w_i c(d, g_i)}{\sum_{i=1}^{k} w_i}}$$

where:

- $c(d, g_i)$ measures how well document d covers requirement g_i.
- w_i represents the importance of requirement g_i.

For example, a document that only says:

Revenue increased by 30%.

has low coverage if the missing information is:

Why did revenue increase?

A document that says:

Revenue increased because demand for AI products grew.

has higher coverage.

9 Term 3: Evidence Novelty

The third component is:

$$N(d, E)$$

It asks:

> Does the candidate document add information that is different from the evidence already collected?

A simple proposed formulation is:

$$\boxed{N(d, E) = 1 - \max_{e \in E} \text{sim}(d, e)}$$

If a candidate is very similar to existing evidence:

$$\text{sim}(d, e) \approx 1$$

then:

$$N(d, E) \approx 0$$

Therefore, the candidate has low novelty.

If the candidate is very different:

$$\text{sim}(d, e) \approx 0$$

then:

$$N(d, E) \approx 1$$

and the candidate has high novelty.

If:

$$E = \emptyset$$

during the first retrieval step, novelty can be treated as maximal or handled separately.

10 The Greek Letters

The Greek letters are simply **weights**.

They control the importance of each part of the retrieval score.

The final score is:

$$S = \alpha R + \beta C + \gamma N$$

where:

$$\alpha = \text{weight of gap relevance}$$
$$\beta = \text{weight of gap coverage}$$
$$\gamma = \text{weight of evidence novelty}$$

A possible constraint is:

$$\boxed{\alpha + \beta + \gamma = 1}$$

with:

$$\alpha, \beta, \gamma \geq 0$$

For example:

$$\alpha = 0.3, \qquad \beta = 0.5, \qquad \gamma = 0.2$$

means that gap coverage is given the largest importance.

The actual values should be determined experimentally or learned using validation data rather than chosen arbitrarily.

11 Complete Mathematical Example

Suppose:

$$R(d, q, G) = 0.8$$

$$C(d, G) = 0.9$$

$$N(d, E) = 0.6$$

and:

$$\alpha = 0.3, \qquad \beta = 0.5, \qquad \gamma = 0.2$$

Then:

$$S(d) = 0.3(0.8) + 0.5(0.9) + 0.2(0.6)$$

Therefore:

$$S(d) = 0.24 + 0.45 + 0.12$$

$$\boxed{S(d) = 0.81}$$

The document receives a GAR score of 0.81.

12 GAR as a Standalone Retriever

GAR should work independently of MIGRES.

A standalone retrieval call can be represented as:

$$(q, E, G, D) \rightarrow GAR \rightarrow TopK\ Documents$$

For a first retrieval:

$$E = \emptyset$$

For an iterative retrieval task, E and G can be updated after every retrieval step.

Therefore, GAR is not a MIGRES-only component.

It can potentially be used in:

- RAG systems,
- multi-hop question answering,
- research systems,
- OSINT systems,
- information-seeking systems,
- other iterative retrieval systems.

13 GAR Inside MIGRES

MIGRES remains our base system.

The original setup is:

$$\boxed{MIGRES + BM25}$$

The proposed setup is:

$$\boxed{MIGRES + GAR}$$

The original retrieval loop can be represented as:

Question → BM25 → Evidence → Missing Information → BM25

Our version becomes:

Question → GAR → Evidence → Missing Information → GAR

The iterative loop belongs to the MIGRES system.

GAR is the retrieval algorithm used at each retrieval step.

14 Experimental Comparison

The main experiment keeps the MIGRES system fixed and changes the retrieval algorithm.

The systems to compare are:

System	**Retriever**
MIGRES + BM25	BM25
MIGRES + DPR	DPR
MIGRES + Contriever	Contriever
MIGRES + ColBERTv2	ColBERTv2
MIGRES + SPLADE	SPLADE
MIGRES + GAR	Proposed method

The original MIGRES configuration acts as the main baseline.

This gives a controlled experiment:

Same system + different retriever

The purpose is to determine whether GAR provides a retrieval advantage.

15 Evaluation Metrics

The project should evaluate both retrieval quality and downstream performance.

15.1 Recall@k

Recall@k asks:

> Did the top k retrieved documents contain the required evidence?

A simple form is:

$$Recall@k = \frac{\text{relevant evidence retrieved in top } k}{\text{total relevant evidence}}$$

15.2 Precision@k

Precision@k asks:

> How much of the retrieved information is useful?

$$Precision@k = \frac{\text{relevant documents in top } k}{k}$$

15.3 MRR

Mean Reciprocal Rank measures how early the first relevant document appears.

$$MRR = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{rank_i}$$

where $rank_i$ is the rank of the first relevant result for query i.

15.4 nDCG

nDCG can be used when documents have graded relevance rather than only relevant/not relevant labels.

15.5 Answer Accuracy

Because the retriever operates inside MIGRES, we can also evaluate whether better retrieval produces better final answers.

15.6 Latency

Measure the time required for retrieval.

15.7 LLM Cost

Measure:

- number of LLM calls,
- input tokens,
- output tokens,
- total retrieval cost.

16 Ablation Studies

An ablation study removes individual parts of GAR.

We can compare:

1. GAR without gap relevance.
2. GAR without gap coverage.
3. GAR without evidence novelty.
4. Full GAR.

The purpose is to answer:

> Does every part of the mathematical model actually contribute?

For example:

$$GAR_{-N}$$

means GAR without the novelty component.

If:

$$GAR > GAR_{-N}$$

on the evaluation metrics, this provides evidence that novelty contributes to the model.

17 Hypothesis Testing

The main hypothesis can be:

Null Hypothesis

$$H_0 :$$

GAR does not significantly improve retrieval performance compared with BM25 when used inside MIGRES.

Alternative Hypothesis

$$H_1 :$$

GAR significantly improves retrieval performance compared with BM25 when used inside MIGRES.

The exact statistical test should be selected after examining the structure of the evaluation data.

We should report:

- statistical significance,
- effect size,
- confidence intervals,
- mean and variance where appropriate.

18 Research Questions

The project can be organized around the following research questions.

RQ1

Does GAR retrieve more useful evidence than BM25?

RQ2

Does GAR outperform strong dense and neural retrieval baselines?

RQ3

Does replacing BM25 with GAR improve the performance of MIGRES?

RQ4

Which component of GAR contributes most to its performance?

RQ5

Does GAR provide these improvements without excessive computational cost?

19 Literature Used

The project uses approximately fifteen papers.

The papers are divided into two groups: papers that motivate/design the retrieval idea and papers used mainly for retrieval comparison and supporting knowledge.

1. **MIGRES — COLING 2025**
   Base system and missing-information idea.
   https://aclanthology.org/2025.coling-main.163/

2. **Unsupervised Alignment-based Iterative Evidence Retrieval — ACL 2020**
   Evidence-gap and uncovered-information retrieval.
   https://aclanthology.org/2020.acl-main.414/

3. **Multi-Hop Paragraph Retrieval — ACL 2019**
   Previous evidence can guide later retrieval.
   https://aclanthology.org/P19-1222/

4. **Better RAG using Relevant Information Gain — 2024**
   Information gain as a retrieval objective.
   https://arxiv.org/abs/2407.12101

5. **InfoGain-RAG — EMNLP 2025**
   Document information gain and usefulness.
   https://aclanthology.org/2025.emnlp-main.365/

6. **Dense Passage Retrieval (DPR) — EMNLP 2020**
   Dense retrieval baseline.
   https://aclanthology.org/2020.emnlp-main.550/

7. **ColBERTv2 — NAACL 2022**
   Late-interaction retrieval baseline.
   https://aclanthology.org/2022.naacl-main.272/

8. **Contriever — EMNLP 2022**
   Unsupervised dense retrieval baseline.
   https://aclanthology.org/2022.emnlp-main.738/

9. **SPLADE — Sparse Neural Retrieval**
   Sparse neural retrieval baseline.
   https://arxiv.org/abs/2107.05720

10. **TART — Task-aware Retrieval with Instructions**
    Task-conditioned retrieval.
    https://arxiv.org/abs/2211.09260

11. **Query2doc — EMNLP 2023**
    Query expansion and richer retrieval representation.
    https://aclanthology.org/2023.emnlp-main.585/

12. **HyDE — ACL 2023**
Hypothetical-document representation for dense retrieval.
https://aclanthology.org/2023.acl-long.99/

13. **EfficientRAG — EMNLP 2024**
Efficient iterative retrieval and reduction of expensive LLM calls.
https://aclanthology.org/2024.emnlp-main.199/

14. **IterCQR — NAACL 2024**
Retrieval-guided query reformulation.
https://aclanthology.org/2024.naacl-long.449/

15. **Adaptive-RAG — NAACL 2024**
Adaptive retrieval strategy selection based on query complexity.
https://aclanthology.org/2024.naacl-long.389/

20 Final Project Architecture

The overall research architecture is:

$$\boxed{\text{Question} \rightarrow \text{MIGRES} \rightarrow \text{Information Gap} \rightarrow \text{GAR} \rightarrow \text{Evidence}}$$

For iterative retrieval:

$$\text{Question} \rightarrow \text{GAR} \rightarrow E_1 \rightarrow G_1 \rightarrow \text{GAR} \rightarrow E_2 \rightarrow G_2 \rightarrow \cdots$$

The important point is that the loop belongs to the surrounding MIGRES system, while GAR is the retrieval algorithm.

GAR can also operate independently:

$$(q, E, G, D) \rightarrow GAR \rightarrow TopK$$

21 Final Research Idea

The final project can be summarized as follows:

> **Gap-Aware Retrieval (GAR) is a proposed retrieval algorithm that uses the original question, the information currently missing, and previously collected evidence to select documents that are relevant to the information gap, cover the missing information, and add non-redundant evidence.**

MIGRES is used as the base system because it already provides an information-gap-driven retrieval setting.

The original system:

$$MIGRES + BM25$$

is compared with:

$$MIGRES + DPR$$

$$MIGRES + Contriever$$

$$MIGRES + ColBERTv2$$

$$MIGRES + SPLADE$$

and finally:

$$\boxed{MIGRES + GAR}$$

The final objective is to determine whether explicitly making the retrieval process **gap-aware** improves evidence retrieval and downstream performance.